What Binds Atoms Together in a Solid?

The forces that hold atoms or ions together in a solid are fundamentally **electromagnetic in nature**. At atomic scale distances, gravitational interactions are negligible, and nuclear forces operate only within the femtometer range of the nucleus. Thus, all cohesive interactions in condensed matter arise from Coulomb electrostatic interactions between charged particles (positively charged atomic nuclei and negatively charged electrons), modified by quantum mechanical phenomena.

When neutral atoms are brought close together to form a solid, their valence electron distributions reconfigure to minimize the total potential energy of the combined system. This electronic reconfiguration manifests primarily through two distinct pathways: either through the complete transfer of valence electrons from one atom to anotheryielding oppositely charged ions that attract electrostaticallyor through the spatial sharing of valence electrons between adjacent atoms to form directional covalent bonds.

1. Ionic Bonding: Energetics and Interionic Potential

Ionic bonding occurs in solids formed from elements with large differences in electronegativity, such as alkali halides. Valence electrons transfer completely from electropositive metal atoms to electronegative non-metal atoms, forming stable closed-shell ions.

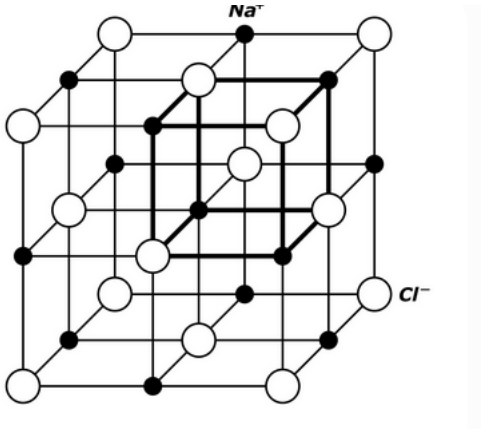


Figure 1: Interpenetrating face-centered cubic (FCC) lattice structure of sodium chloride (NaCl).

A. Formation of Ions: Energetics of NaCl

To understand the energy gain during ionic solid formation, consider creating a single pair of Na^+ and Cl^- ions from isolated neutral atoms:

1. **Ionization Energy (I):** Energy required to strip a valence electron from a neutral sodium atom:
$$\mathrm{Na} \rightarrow \mathrm{Na}^+ + e^-, \quad I = +5.14 \text{ eV}$$
2. **Electron Affinity (EA):** Energy released when a neutral chlorine atom accepts an electron:
$$\mathrm{Cl} + e^- \rightarrow \mathrm{Cl}^-, \quad EA = -3.61 \text{ eV}$$

Creating isolated Na^+ and Cl^- ions in a vacuum requires a net energy input:

$$\Delta E_{\text{isolated}} = I - EA = +5.14 \text{ eV} - 3.61 \text{ eV} = +1.53 \text{ eV}$$

When the two ions approach each other to an equilibrium bond distance $R_0 \approx 2.81$ Å, electrostatic attraction lowers the potential energy:

$$\Delta U_{\text{Coulomb}}(R_0) = -\frac{e^2}{4\pi\varepsilon_0 R_0} \approx -5.12 \text{ eV}$$

The net energy change for a bound pair is:

$$\Delta E_{\text{pair}} = I - EA - \frac{e^2}{4\pi\varepsilon_0 R_0} \approx +1.53 \text{ eV} - 5.12 \text{ eV} = -3.59 \text{ eV}$$

B. Short-Distance Repulsion Mechanisms

At short internuclear distances, two physical mechanisms create a sharp repulsive core:

1. **Incomplete Screening & Nuclear Repulsion:** At very short distances ($R \ll R_0$), overlapping electron clouds expose unshielded positive nuclei to direct Coulomb repulsion ($1/R$).
2. **Pauli Exclusion Principle:** Overlapping filled electron shells force electrons into higher unoccupied quantum states, producing a steep repulsive force.

C. Total Effective Potential Energy Curve

Combining electrostatic attraction with short-range Pauli repulsion gives the total interionic potential energy (Born-Landé potential):

$$U(R) = -\frac{\alpha e^2}{4\pi\varepsilon_0 R} + \frac{B}{R^n}$$

Where α is the Madelung constant ($\alpha \approx 1.74756$ for NaCl) and B/R^n models Pauli core repulsion ($n \approx 6 - 12$).

Minimizing $U(R)$ at R_0 yields the cohesive lattice energy per ion pair:

$$U_{\text{cohesive}}(R_0) = -\frac{\alpha e^2}{4\pi\varepsilon_0 R_0}\left(1 - \frac{1}{n}\right)$$

2. Covalent Bonding

Covalent bonding is considerably more subtle and delicate in nature than ionic bonding. Rather than involving a complete transfer of electric charge, covalent bonding arises from the **spatial sharing and delocalization of electrons** between adjacent atomic nuclei.

When two atoms approach each other, their atomic orbital wavefunctions overlap. The electron cloud is no longer localized around a single nucleus; instead, it delocalizes over both nuclei. This increased spatial spreading lowers the electron's kinetic energy while increasing the electron density directly between the positively charged nuclei, screening their mutual repulsion and producing a net attractive force.

A. The Simplest Covalent System: The H_2^+ Ion

To understand the quantum mechanics of covalent bonding, we consider the simplest possible system: the ionized hydrogen molecule (H_2^+), consisting of two protons (A and B) separated by internuclear distance R, and a single electron.

1. The Hamiltonian In the Born-Oppenheimer approximation, the non-relativistic Hamiltonian for the electron is:

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 - \frac{e^2}{4\pi\varepsilon_0 r_A} - \frac{e^2}{4\pi\varepsilon_0 r_B} + \frac{e^2}{4\pi\varepsilon_0 R}$$

where $r_A = |\mathbf{r} - \mathbf{R}_A|$ and $r_B = |\mathbf{r} - \mathbf{R}_B|$.

2. The Schrödinger Equation

$$\hat{H}\psi(\mathbf{r}) = E\psi(\mathbf{r})$$

B. Variational Calculation and Trial Wavefunction

According to the Variational Principle:

$$E[\psi_{\text{trial}}] = \frac{\langle\psi_{\text{trial}}|\hat{H}|\psi_{\text{trial}}\rangle}{\langle\psi_{\text{trial}}|\psi_{\text{trial}}\rangle} \geq E_0$$

We construct our trial wavefunction using a Linear Combination of Atomic Orbitals (LCAO) centered on each proton:

$$\psi_{\text{trial}}(\mathbf{r}) = C_1\phi_A(\mathbf{r}) + C_2\phi_B(\mathbf{r})$$

where ϕ_A and ϕ_B are hydrogen $1s$ ground state atomic orbitals (a_0 is the Bohr radius):

$$\phi_A(\mathbf{r}) = \frac{1}{\sqrt{\pi a_0^3}}e^{-r_A/a_0}, \quad \phi_B(\mathbf{r}) = \frac{1}{\sqrt{\pi a_0^3}}e^{-r_B/a_0}$$

These orbitals satisfy single-atom Schrödinger equations:

$$\left(-\frac{\hbar^2}{2m}\nabla^2 - \frac{e^2}{4\pi\varepsilon_0 r_A}\right)\phi_A = E_1\phi_A$$

with $E_1 = -13.6$ eV.

C. Energy Expectation Value with Complex Coefficients

In general, the expansion coefficients C_1 and C_2 in the trial wavefunction can be complex numbers. The norm of the trial state is given by:

$$\begin{aligned}\langle\psi_{\text{trial}}|\psi_{\text{trial}}\rangle &= \int (C_1^*\phi_A^* + C_2^*\phi_B^*)(C_1\phi_A + C_2\phi_B)\, d^3r \\ &= |C_1|^2 S_{AA} + |C_2|^2 S_{BB} + C_1^* C_2 S_{AB} + C_2^* C_1 S_{BA}\end{aligned}$$

Assuming normalized, real atomic orbitals ($\phi_A^* = \phi_A$, $\phi_B^* = \phi_B$, $S_{AA} = S_{BB} = 1$) and spatial symmetry ($S_{AB} = S_{BA} = S$), this simplifies to:

$$\langle\psi_{\text{trial}}|\psi_{\text{trial}}\rangle = |C_1|^2 + |C_2|^2 + (C_1^*C_2 + C_2^*C_1)S$$

where $S(R)$ is the spatial overlap integral:

$$S(R) = \langle\phi_A|\phi_B\rangle = \int \phi_A^*(\mathbf{r})\phi_B(\mathbf{r})\, d^3r = e^{-R/a_0}\left(1 + \frac{R}{a_0} + \frac{1}{3}\left(\frac{R}{a_0}\right)^2\right)$$

Similarly, the Hamiltonian expectation value is:

$$\langle\psi_{\text{trial}}|\hat{H}|\psi_{\text{trial}}\rangle = |C_1|^2H_{AA} + |C_2|^2H_{BB} + C_1^*C_2H_{AB} + C_2^*C_1H_{BA}$$

Decomposing $\hat{H} = \hat{H}_A - \frac{e^2}{4\pi\varepsilon_0 r_B} + \frac{e^2}{4\pi\varepsilon_0 R}$, we define the diagonal and off-diagonal matrix elements:

1. **Diagonal Matrix Element ($H_{AA} = H_{BB} \equiv H_0$):**

$$H_0 = \langle\phi_A|\hat{H}|\phi_A\rangle = E_1 + \frac{e^2}{4\pi\varepsilon_0 R} - J(R)$$

where $J(R)$ is the classical Coulomb integral:

$$J(R) = \int |\phi_A(\mathbf{r})|^2 \frac{e^2}{4\pi\varepsilon_0 r_B}\, d^3r = \frac{e^2}{4\pi\varepsilon_0 R}\left[1 - \left(1 + \frac{R}{a_0}\right)e^{-2R/a_0}\right]$$

2. **Off-Diagonal Matrix Element ($H_{AB} = H_{BA} \equiv H_{12}$):**

$$H_{12} = \langle\phi_A|\hat{H}|\phi_B\rangle = \left(E_1 + \frac{e^2}{4\pi\varepsilon_0 R}\right)S(R) - K(R)$$

where $K(R)$ is the quantum mechanical exchange integral:

$$K(R) = \int \phi_A^*(\mathbf{r})\frac{e^2}{4\pi\varepsilon_0 r_B}\phi_B(\mathbf{r})\, d^3r = \frac{e^2}{4\pi\varepsilon_0 a_0}\left(1 + \frac{R}{a_0}\right)e^{-R/a_0}$$

Thus, the full energy functional as a function of the complex coefficients C_1, C_2, C_1^*, C_2^* is:

$$E(C_1, C_2, C_1^*, C_2^*) = \frac{|C_1|^2H_0 + |C_2|^2H_0 + (C_1^*C_2 + C_2^*C_1)H_{12}}{|C_1|^2 + |C_2|^2 + (C_1^*C_2 + C_2^*C_1)S}$$

Writing this in implicit product form gives:

$$E\left[|C_1|^2 + |C_2|^2 + (C_1^*C_2 + C_2^*C_1)S\right] = |C_1|^2H_0 + |C_2|^2H_0 + (C_1^*C_2 + C_2^*C_1)H_{12} \quad (1)$$

D. Variational Derivation, Secular Equations, and Two-State Analogy

Treating C_k and C_k^* as independent variables in quantum mechanics, we minimize E with respect to the complex conjugates C_1^* and C_2^*:

$$\frac{\partial E}{\partial C_1^*} = 0 \quad \text{and} \quad \frac{\partial E}{\partial C_2^*} = 0$$

Differentiation with respect to C_1^*: Differentiating Eq. (1) with respect to C_1^*:

$$\frac{\partial E}{\partial C_1^*}\left[|C_1|^2 + |C_2|^2 + (C_1^* C_2 + C_2^* C_1)S\right] + E(C_1 + C_2 S) = C_1 H_0 + C_2 H_{12}$$

Setting $\frac{\partial E}{\partial C_1^*} = 0$:

$$C_1(H_0 - E) + C_2(H_{12} - ES) = 0 \tag{2}$$

Differentiation with respect to C_2^*: Similarly, differentiating Eq. (1) with respect to C_2^* and setting $\frac{\partial E}{\partial C_2^*} = 0$:

$$C_1(H_{12} - ES) + C_2(H_0 - E) = 0 \tag{3}$$

Matrix Representation and Analogy to a Two-State System Equations (2) and (3) form a generalized matrix eigenvalue problem:

$$\begin{pmatrix} H_0 & H_{12} \\ H_{12} & H_0 \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E \begin{pmatrix} 1 & S \\ S & 1 \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} \tag{4}$$

If the spatial overlap $S \ll 1$ is neglected ($S \to 0$), the matrix equation simplifies to the canonical **two-level quantum system**:

$$\begin{pmatrix} H_0 & H_{12} \\ H_{12} & H_0 \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} \tag{5}$$

This is structurally identical to two-state quantum systems like a spin-1/2 particle in a magnetic field or ammonia inversion (NH_3). Here, H_0 represents the localized state energy on either nucleus, while H_{12} is the off-diagonal tunneling/coupling matrix element that breaks the degeneracy.

Solving the secular determinant $\det(\mathbf{H} - E\mathbf{S}) = 0$ yields two physical state solutions:

1. **Bonding State (ψ_+):** Symmetric state with $C_1 = C_2 = \frac{1}{\sqrt{2(1+S)}}$

$$\psi_+(\mathbf{r}) = \frac{1}{\sqrt{2(1+S)}}(\phi_A + \phi_B)$$

$$E_+(R) = \frac{H_0 + H_{12}}{1 + S(R)} = E_1 + \frac{e^2}{4\pi\varepsilon_0 R} - \frac{J(R) + K(R)}{1 + S(R)}$$

2. **Anti-Bonding State (ψ_-):** Antisymmetric state with $C_1 = -C_2 = \frac{1}{\sqrt{2(1-S)}}$

$$\psi_-(\mathbf{r}) = \frac{1}{\sqrt{2(1-S)}}(\phi_A - \phi_B)$$

$$E_-(R) = \frac{H_0 - H_{12}}{1 - S(R)} = E_1 + \frac{e^2}{4\pi\varepsilon_0 R} - \frac{J(R) - K(R)}{1 - S(R)}$$

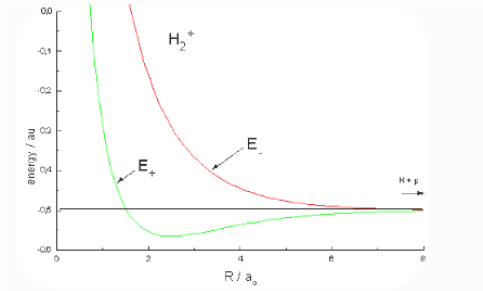


Figure 2: Variational potential energy curves $E_+(R)$ (bonding) and $E_-(R)$ (anti-bonding) for H_2^+ as a function of internuclear separation R.

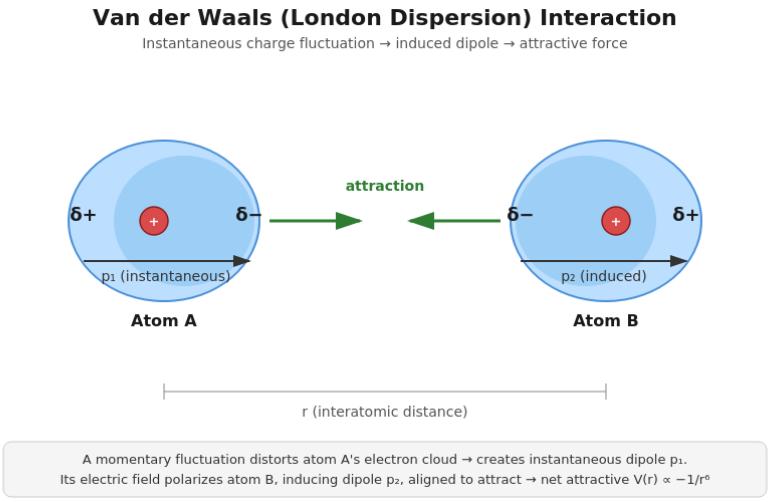


Figure 3: Schematic of van der Waals interaction between two neutral atoms A and B separated by internuclear distance R. An instantaneous fluctuating electric dipole $\mathbf{p}_A$ on Atom A creates an electric field $\mathbf{E}_A$, inducing a secondary electric dipole $\mathbf{p}_B$ on Atom B.

3. Van der Waals Bonding: Quantum Dipole Fluctuations

Van der Waals interactions represent a weaker, non-directional cohesive mechanism present between all neutral atoms and molecules. Unlike ionic or covalent bonds, van der Waals forces do not rely on a net ionic charge or static electron sharing. Instead, they arise from instantaneous **quantum fluctuations** in the electron charge density of spherically symmetric atoms.

A. Instantaneous Dipoles and Polarization Field

Consider two neutral, isolated, spherically symmetric atoms (A and B) separated by internuclear distance R. On time average, the negative electron cloud is centered symmetrically around the positive nucleus, leading to zero net static electric dipole moment:

$$\langle \mathbf{p}_A \rangle = 0, \quad \langle \mathbf{p}_B \rangle = 0$$

However, due to quantum mechanical uncertainty and zero-point fluctuations, the instantaneous position of an electron relative to its nucleus constantly varies. At any given instant t, atom A exhibits a non-zero **instantaneous electric dipole moment**:

$$\mathbf{p}_A = e\mathbf{r}_A$$

where $\mathbf{r}_A$ is the instantaneous displacement vector between the nucleus and the electron cloud center of atom A.

This fluctuating dipole produces an electric field $\mathbf{E}_A$ at the site of atom B (separated by distance R along the axis):

$$E_A(R) \sim \frac{1}{4\pi\varepsilon_0}\frac{p_A}{R^3}$$

The electric field $\mathbf{E}_A$ polarizes the neutral electron cloud of atom B, inducing a secondary electric dipole moment $\mathbf{p}_B$:

$$\mathbf{p}_B = \alpha_B \mathbf{E}_A \sim \frac{\alpha_B}{4\pi\varepsilon_0}\frac{\mathbf{p}_A}{R^3}$$

where α_B represents the static atomic polarizability of atom B.

B. Interaction Potential Energy Derivation

The electrostatic dipole-dipole interaction potential energy $U(R)$ between the instantaneous dipole $\mathbf{p}_A$ and the induced dipole $\mathbf{p}_B$ is given by:

$$U(R) = -\mathbf{p}_B \cdot \mathbf{E}_A \sim -\frac{1}{4\pi\varepsilon_0}\frac{p_A p_B}{R^3}$$

Substituting the expression for the induced dipole $\mathbf{p}_B$:

$$U(R) \sim -\frac{1}{4\pi\varepsilon_0}\frac{p_A\left(\frac{\alpha_B}{4\pi\varepsilon_0}\frac{p_A}{R^3}\right)}{R^3} = -\frac{\alpha_B}{(4\pi\varepsilon_0)^2}\frac{p_A^2}{R^6}$$

Taking the quantum mechanical expectation value (ground-state time average) yields:

$$\langle U(R) \rangle \sim -\frac{\alpha_B}{(4\pi\varepsilon_0)^2}\frac{\langle p_A^2 \rangle}{R^6}$$

Key Physical Insights:

1. **Universal Attraction:** Even though the time-averaged dipole moment vanishes ($\langle \mathbf{p}_A \rangle = 0$), the mean-squared dipole expectation value is strictly positive ($\langle p_A^2 \rangle \equiv \langle |\mathbf{p}_A|^2 \rangle > 0$). Consequently, $\langle U(R) \rangle < 0$, guaranteeing that London dispersion forces are **universally attractive**.

2. R^{-6} **Distance Scaling:** The potential decays as $1/R^6$, which reflects the product of electric field generation ($\sim R^{-3}$) and induced dipole potential energy ($\sim R^{-3}$).

3. **London Formula:** A full quantum perturbation calculation for two identical atoms gives the explicit London dispersion energy:

$$U_{\text{dispersion}}(R) = -\frac{3}{2}\frac{I\alpha^2}{(4\pi\varepsilon_0)^2 R^6} \equiv -\frac{C_6}{R^6}$$

   where I is the characteristic atomic ionization energy.